6.3.7 *Fokker-Planck Equation for Systems of SDEs*

We describe the generalization of the Itō-Doeblin formula and the Fokker-Planck equation to a system of SDEs. If ϕ is a scalar valued function of $\mathbb{R}^m$, and $\mathbf{Z}(t)$ is the solution of the SDE system

$$\mathrm{d}\mathbf{Z} = \boldsymbol{a}(\mathbf{Z})\mathrm{d}t + \boldsymbol{B}\,\mathrm{d}\mathbf{W}, \tag{6.39}$$

where $\boldsymbol{a}$ is a function in $\mathbb{R}^m$ and $\boldsymbol{B}$ is a constant matrix, then we have

$$\mathrm{d}\phi(\mathbf{Z}(t)) = \nabla\phi \cdot [\boldsymbol{a}\mathrm{d}t + \boldsymbol{B}\mathrm{d}\mathbf{W}] + \frac{1}{2}\mathrm{d}\mathbf{W}^T\boldsymbol{B}^T\phi''\boldsymbol{B}\mathrm{d}\mathbf{W}, \tag{6.40}$$

where ϕ'' is the Hessian matrix of ϕ, i.e. the matrix of second partial derivatives of ϕ with respect to pairs of the variables. In some applications $\boldsymbol{B} = \boldsymbol{B}(\mathbf{Z})$, in which case the SDE (6.39) is said to involve *multiplicative noise* (as oppose to the *additive* case we consider when $\boldsymbol{B}$ is constant). For a discussion of the differences and applications of these types of noise, see [289, 324].

All components on the right hand side of (6.40) are evaluated at $\mathbf{Z}(t)$. Then using the shorthand rules $\mathrm{dW}_i\mathrm{dW}_j = \mathrm{d}t\delta_{ij}$ (where δ_{ij} is one if $i = j$, zero otherwise), we have

$$\mathrm{d}\phi(\mathbf{Z}(t)) = \nabla\phi \cdot [\boldsymbol{a}\mathrm{d}t + \boldsymbol{B}\mathrm{d}\mathbf{W}] + \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{B}^T\phi''\boldsymbol{B}\right)\mathrm{d}t, \tag{6.41}$$

where $\mathrm{Tr}(\cdot)$ refers to the matrix trace, the sum of the diagonal elements of the given matrix.

In the multivariate case, the density is a function of the vector describing a state of the system z and t, and is governed by the partial differential equation

$$\frac{\partial\rho}{\partial t} = -\nabla \cdot (\boldsymbol{a}(z)\rho(z)) + \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{B}^T\rho''\boldsymbol{B}\right).$$

The second order "diffusion" term in this expression may be expanded as

$$\mathrm{Tr}\left(\boldsymbol{B}^T\rho''\boldsymbol{B}\right) = \sum_{i=1}^{m}\sum_{j=1}^{m}\left[\boldsymbol{B}\boldsymbol{B}^T\right]_{ij}\frac{\partial^2\rho}{\partial z_i\partial z_j}.$$